Knowledge Unlearning in Language Models using Sparse Autoencoders

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Abstract

Large Language Models (LLMs) often inadvertently memorize copyrighted intellectual property (IP) during pre-training. Current state-of-the-art methods for unlearning, such as those proposed by Eldan and Russinovich [4], rely on "approximate unlearning" via black-box fine-tuning that modifies model weights globally. In this work, we present a mechanistic approach that aims to surpass these methods with precision and interpretability. By leveraging Sparse Autoencoders (SAEs) on the residual stream of GPT-2 Medium (Layer 12), we identify and ablate specific interpretable features associated with targeted IP specifically the Harry Potter universe. Our implementation achieves a **77% reduction** in Harry Potter (referred as HP in this report) knowledge recall with negligible impact on general model capabilities (WikiText-2 perplexity degradation of only 0.1%), providing a more interpretable and precise alternative to gradient-based unlearning. We conduct a rigorous head-to-head comparison against the Eldan & Russinovich baseline, demonstrating competitive unlearning depth with improved model stability. The code for this project is available in this GitHub repository. Pretrained Model and other resources are available at this drive link.

1 Introduction

The ethical and legal implications of Large Language Models (LLMs) training on copyrighted data have reached a critical inflection point. Landmark works such as Eldan and Russinovich [4] have demonstrated that models like Llama-2 can be fine-tuned to forget specific content by replacing idiosyncratic terms with generic alternatives and steering the model's predictions away from domain-specific completions. However, this method modifies the model's weights globally through fine-tuning, offering limited transparency into *how* the knowledge is actually excised and risking collateral damage to related capabilities.

We propose a fundamentally different paradigm: **mechanistic unlearning via Sparse Autoencoders (SAEs)**. Rather than re-training or fine-tuning the model, we identify the specific directions in the residual stream that encode IP-related concepts, then ablate those features at inference time. This approach is grounded in the Mechanistic Interpretability literature and offers three key advantages over prior work:

- **Interpretability**: The ablated features are human-inspectable; we can verify *which* concepts are being removed.
- **No weight modification**: Unlearning is applied as inference-time hooks, leaving the base model intact and making the process reversible.
- **High Precision**: By targeting features identified through Difference-in-Means analysis, we minimize damage to adjacent capabilities such as general magical / fantasy comprehension or language fluency.

We evaluate our method on GPT-2 Medium, using the Harry Potter series as the unlearn target, and directly compare our results against the Eldan & Russinovich baseline across completion recall, log-probability shifts, and standard NLP benchmarks. The remainder of this paper is organized as follows: Section 2 reviews related work, Section 4 describes our methodology, Section 5 details the experimental setup, Section 6 presents our results and comparison, and Section 9 concludes.

2 Related Work

Machine Unlearning. Machine unlearning was originally motivated by privacy concerns and the right of individuals to request their data be removed from trained models [2]. Early work focused primarily on classification models where exact unlearning is tractable by retraining on a reduced dataset. For large generative models, exact unlearning is prohibitively expensive given the scale of modern pretraining.

Approximate Unlearning in LLMs. Eldan and Russinovich [4] introduced the first effective technique for knowledge unlearning in generative LLMs. Their three-component approach reinforcement bootstrapping, anchored term replacement, and generic label fine-tuning demonstrated that Llama-2-7B could be made to forget the Harry Potter series in approximately one GPU-hour. Critically, benchmark performance (ARC, BoolQ, HellaSwag, PIQA, WinoGrande) was largely preserved. However, the method is fundamentally a black-box

weight-editing procedure with no guarantee of which internal representations are modified.

Mechanistic Interpretability & SAEs. Recent advances in Mechanistic Interpretability, particularly the development of Sparse Autoencoders as a tool for decomposing model activations into interpretable features [1], have opened a new avenue for targeted model editing. SAEs learn a sparse, over-complete basis for neural activations, where individual basis vectors often correspond to human-interpretable concepts. This is precisely the handle we exploit for IP unlearning.

Model Editing. Parallel work on model editing [7] has shown that specific factual associations can be traced to specific MLP layers and overwritten. Our approach differs in that we do not edit weights but instead suppress activation directions, and we target a broad semantic domain rather than isolated facts.

3 Background: Sparse Autoencoders for Interpretability

At the core of our approach is the Sparse Autoencoder (SAE), an unsupervised neural network architecture designed to learn disentangled and semantically meaningful features from complex data. While traditional autoencoders are primarily used for dimensionality reduction, SAEs are optimized for creating overcomplete, interpretable feature dictionaries from the internal activations of large language models (LLMs).

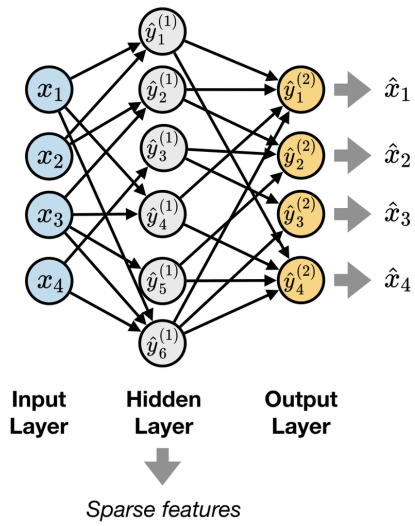
3.1 Sparsity

A standard autoencoder consists of two components: an **encoder**, f, which maps an input vector $\mathbf{x} \in R^d$ to a latent representation $\mathbf{z} \in R^m$, and a **decoder**, g, which reconstructs the input from this latent code, $\hat{\mathbf{x}} = g(f(\mathbf{x}))$. The model is trained to minimize the reconstruction error, typically the Mean Squared Error (MSE):

$$\mathcal{L}_{\text{recon}} = \|\mathbf{x} - \hat{\mathbf{x}}\|_2^2 \tag{1}$$

Historically, autoencoders forced the model to learn salient features by creating a dimensional bottleneck where $m \ll d$.

Sparse Autoencoders depart from this paradigm. Instead of a dimensional bottleneck, they often employ an **overcomplete** latent space where the feature dimension m is significantly larger than the input dimension d (i.e., $m \gg d$). To prevent the model from learning a trivial identity function, a **sparsity constraint** is imposed on the latent activations $\mathbf{z}$. This constraint encourages the network to represent each input $\mathbf{x}$ using only a small subset of the available features in the latent dictionary (32 in our case).



The motivation for this design comes from the **sparse coding hypothesis**, which states that natural signals can be represented using only a few components from a larger set of basic elements . In large language models, this means that dense activation vectors may actually be combinations of a small number of underlying features. Sparse autoencoders try to uncover these features by breaking the model's mixed activations into sparse, more interpretable components, sometimes referred to as monosemantic or disentangled features \[}**?**].

3.2 Enforcing Sparsity

Sparsity can be enforced in several ways. The classic approach involves adding an L_1 regularization term to the loss function, which penalizes the sum of the absolute values of the latent activations:

$$\mathcal{L} = \|\mathbf{x} - \hat{\mathbf{x}}\|_2^2 + \lambda \|\mathbf{z}\|_1 \tag{2}$$

The hyperparameter λ controls the trade-off between reconstruction fidelity and the degree of sparsity.

However, tuning λ is challenging and often requires carefully balancing reconstruction accuracy and feature sparsity. A more direct approach, which we use in our method, is the \textbf{Top-K activation function} \[5]. This function keeps only the k largest activations in $\mathbf{z}$ and sets all others to zero during the forward pass. As a result, each input has a fixed level of sparsity without relying on a sensitive regularization parameter in the loss function. This effectively separates the goals of reconstruction and sparsity.

By training such an SAE on the internal activations of an LLM, we obtain a dictionary of features where each feature vector (a column of the decoder matrix) ideally corresponds to a single, human-interpretable concept. This dictionary then serves as the basis for our knowledge erasure intervention, described in Section~**??**{sec:method}.

4 Methodology

Our proposed approach selectively eradicates domain-specific knowledge from GPT-2 Medium (355M parameters) without altering the model's foundational weights. For our implementation [1], the methodology is structured as a comprehensive five-stage pipeline: data corpus construction, Sparse Autoencoder (SAE) training, feature identification, inference-time intervention, and rigorous evaluation.

4.1 Stage 1: Data Corpus Construction

To accurately isolate features corresponding to the target domain without inducing catastrophic forgetting of general linguistic or thematic capabilities, we construct two contrasting corpora:

- **Target Corpus (Harry Potter)**: The complete Harry Potter book series is aggregated and heavily preprocessed to extract text segments with a high density of domain-specific knowledge tokens (e.g., character names, spells, locations). Passages are tokenized and chunked into 512-token blocks using the standard GPT-2 tokenizer. The target corpus we used has **1,307,107** tokens.

- **Neutral Baseline Corpus**: A diverse mixture comprising WikiText-2 for general encyclopedic syntax and TinyStories for foundational narrative structures. The token count and sampling ratio is as follows

Dataset	Tokens	Sampling Ratio
WikiText-2	2,391,884	0.851
TinyStories	417,455	0.149
Total	2,809,339	1.000

Table 1: Composition of the neutral corpus used for baseline language modeling statistics.

The deliberate inclusion of general fiction (TinyStories) alongside factual text (WikiText-2) in the neutral corpus is a critical design choice. It establishes a broadened baseline that prevents the SAE from erroneously entangling general high-fantasy concepts (e.g., "wand", "wizard", "dragon") with Harry-Potter-specific features.

4.2 Stage 2: Top-K Sparse Autoencoder (SAE) Training

The core of our pipeline is training a Top-K Sparse Autoencoder to learn an interpretable, overcomplete feature dictionary of the residual stream. Based on preliminary probing experiments, which show that high-level semantic concepts and fictional world knowledge are more linearly separable in deeper layers, we focus on Layer 12 of GPT-2 Medium ($d_{\text{model}} = 1024$).

[1] https://github.com/Jay-G14/INLP_PROJECT/

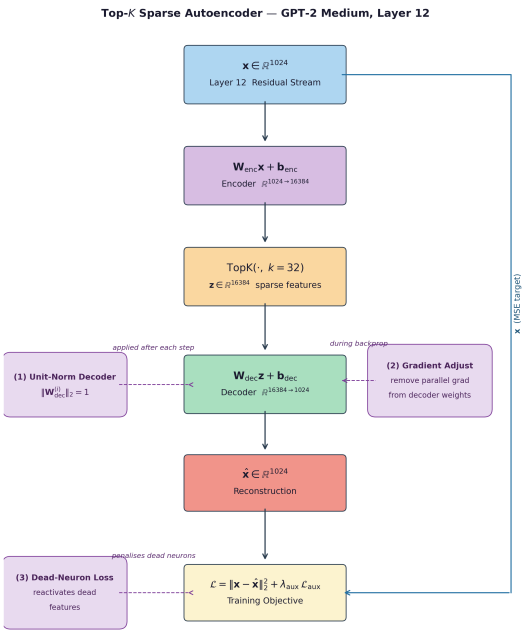


Figure 1: Top-K Sparse Autoencoder (SAE) training architecture

Given an activation vector $\mathbf{x} \in R^d$, the SAE projects it into a higher-dimensional feature space and reconstructs it as $\hat{\mathbf{x}}$:

$$\mathbf{z} = \text{TopK}(\mathbf{W}_{\text{enc}}\mathbf{x} + \mathbf{b}_{\text{enc}}, k) \tag{3}$$

$$\hat{\mathbf{x}} = \mathbf{W}_{\text{dec}}\mathbf{z} + \mathbf{b}_{\text{dec}} \tag{4}$$

We employ an expansion factor of 16, mapping the 1024-dimensional residual stream to a feature dictionary of 16,384 distinct features. The TopK activation function explicitly enforces sparsity by retaining only the $k = 32$ largest activations and zeroing out the rest, natively circumventing the shrinkage issues associated with L_1 regularization.

To ensure stable dictionary learning, our training loop includes three key regularization techniques:

1. **Unit-Norm Decoder Constraint**: Decoder columns are normalized so that $\|\mathbf{W}_{\text{dec}}^{(i)}\|_2 = 1$. This prevents the model from trivially reducing the loss by shrinking $\mathbf{z}$ while increasing $\mathbf{W}_{\text{dec}}$.

2. **Gradient Adjustment for Decoder Weights**: During backpropagation, we modify the decoder gradients to prevent a few dominant features from overpowering newly emerging ones.

3. **Auxiliary Dead-Neuron Loss**: To use the dictionary capacity effectively, an auxiliary loss $\mathcal{L}_{\text{aux}}$ is added to reactivate "dead" neurons (features that rarely activate across large batches).

The final objective minimized during training is the reconstruction Mean Squared Error (MSE) coupled with the dead-neuron penalty:

$$\mathcal{L} = \|\mathbf{x} - \hat{\mathbf{x}}\|_2^2 + \lambda_{\text{aux}}\mathcal{L}_{\text{aux}} \tag{5}$$

4.3 Stage 3: Feature Identification via Difference-in-Means

Post-training, we perform a difference-in-means analysis to isolate the subset of the 16,384 features that strictly encode Harry Potter knowledge. For each feature i, we compute a **Specificity Ratio** R_i:

$$R_i = \frac{\mu_i^{\text{target}} + \epsilon}{\mu_i^{\text{neutral}} + \epsilon} \tag{6}$$

where μ_i^{target} and μ_i^{neutral} denote the mean activation values of feature i across the target and neutral corpora, respectively, and ϵ is a minimal smoothing scalar.

Features exhibiting a specificity ratio $R_i \geq 20$ (i.e., firing at least 20 times more frequently on Harry Potter text than on the neutral baseline) are flagged as domain-specific. From this flagged subset, we select the top 100 highest-magnitude features to form our intervention set $\mathcal{F}$.

4.4 Stage 4: Inference-Time Intervention

We seamlessly integrate the knowledge ablation into the model's forward pass using `TransformerLens` . By registering forward hooks at Layer 12, we intercept the residual stream activations prior to the next attention block.

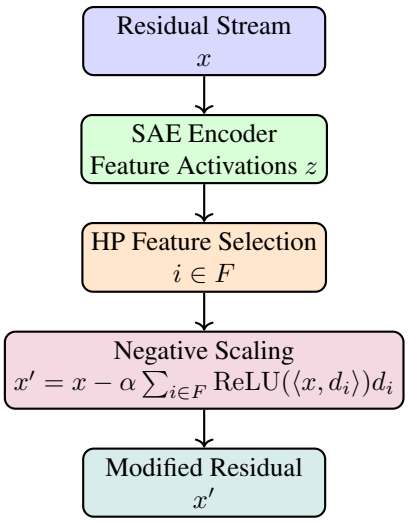


Figure 2: Inference-time intervention pipeline. Residual activations are projected into SAE features, Harry-Potter-specific features are identified, and negative scaling suppresses these directions before continuing the forward pass.

For each feature $i \in \mathcal{F}$ with corresponding decoder direction $\mathbf{d}_i \in R^d$, we apply a **conditional negative scaling** intervention:

$$\mathbf{x}' = \mathbf{x} - \alpha \sum_{i \in \mathcal{F}} \text{ReLU}\big(\langle \mathbf{x}, \mathbf{d}_i \rangle\big) \mathbf{d}_i \tag{7}$$

where $\alpha = -3.0$ dictates the ablation magnitude, empirically tuned on a held-out validation set. Unlike standard ablation which simply zeroes out activations, negative scaling actively suppresses these geometric directions when the model attempts to utilize them.

Importantly, this intervention only affects the small subspace defined by $\mathcal{F}$. The rest of the activation vector remains unchanged, so the overall **reconstruction quality is preserved**. As a result, the base LLM maintains its grammatical fluency and general reasoning ability without requiring any weight retraining.

To complement the spatial ablation, we apply Top-p sampling ($p = 0.9$) and a frequency penalty ($\lambda_f = 1.0$) during the auto-regressive decoding phase to further mitigate the likelihood of repetitive, localized domain hallucinations.

5 Experimental Setup & Evaluation

We now evaluate the effectiveness of the ablation. We build a benchmark to compare our inference-time intervention with fine-tuning based unlearning methods.

5.1 Models Evaluated

- **Ours (SAE-Ablated)**: GPT-2 Medium initialized with standard pre-trained weights, augmented dynamically via SAE ablation hooks at Layer 12 ($\alpha = -3.0$).
- **Baseline (Eldan & Russinovich, 2023)**: Llama-2-7B, subjected to reinforcement bootstrapping and anchored term replacement as detailed in Eldan and Russinovich [4].

Note: The baseline model uses a different architecture (GPT-2 vs. Llama-2-7B). We use GPT-2 due to computational constraints, which make training and experimentation with larger models infeasible in our setting. While this prevents a direct parameter-level comparison, the baseline is included to provide a behavioral reference point for commonly used unlearning evaluation metrics.

5.2 Evaluation Metrics

We evaluate the models on two aspects: the removal of the targeted knowledge, and the preservation of the model's general capabilities.

HP Knowledge Eradication

- **HP Recall (LLM-as-Judge)**: We query the model with 300 domain-specific completion prompts, meticulously generated following the elicitation methodology of Eldan and Russinovich [4]. Output completions are fed to an LLM judge (`moonshotai/kimi-k2-instruct`), which categorizes the residual knowledge into four distinct familiarity tiers (e.g., from "No Knowledge" to "Exact Memorization").
- **Domain-Specific Log-Probability**: For a curated subset of prompts, we directly measure the negative log-likelihood the model assigns to ground-truth Harry Potter tokens, quantifying the suppression at the distributional level.

Metric	Baseline (Top-P)	Ablated (Scale -3.0)	Impact (Δ)
Target Knowledge			
HP Log-Probability	-3.6567	-5.1248	$\mathbf{-1.4681}$ **(Massive Drop)**
Surgical Selectivity			
Magic Log-Probability	-3.0521	-3.1354	Preserved (-0.08)
Fantasy Log-Probability	-6.4321	-6.8215	Minor Collateral (-0.39)
Real World Log-Prob	-3.9647	-3.9692	Unchanged (≈ 0.00)
Capability Stability			
Degeneration Rate	0.0%	0.0%	Perfectly Stable
WikiText-2 Perplexity	43.84	43.88	0.1% Degradation

Table 2: Quantitative evaluation metrics. Log-probabilities represent the model's likelihood of predicting domain-correct tokens. A more negative value in the HP domain indicates successful erasure.

- **Cross-Domain Spillover Analysis**: We probe the log-probabilities of adjacent semantic domains (Magic, General Fantasy, and Real-World facts) to empirically verify that collateral suppression is minimal.

General Capability Preservation

- **WikiText-2 Perplexity**: Measured across held-out corpora to guarantee that language modeling quality on general textual distributions does not degrade post-intervention.
- **Degeneration Rate**: We track the proportion of generated sequences that devolve into token repetition or structural incoherence, ensuring that the negative scaling ($\alpha = -3.0$) does not systematically fracture the residual stream.

Manual Evaluation To complement automated metrics, we conduct a small-scale manual evaluation on 30 Harry Potter prompts. For each prompt, we compare completions produced by the baseline model and the ablated model. A human evaluator with strong domain familiarity (Jayant Gupta, who has read the full Harry Potter series and watched the films multiple times) annotates whether the ablation successfully removes recognizable Harry Potter knowledge.

Each example receives a binary label (`ablation_worked` or `ablation_failed`). An ablation is considered successful if the ablated completion no longer contains identifiable Harry Potter references while the baseline completion clearly reflects such knowledge. We report the percentage of prompts exhibiting Harry Potter knowledge in the baseline and the percentage for which the ablation successfully suppresses it.

6 Results and Analysis

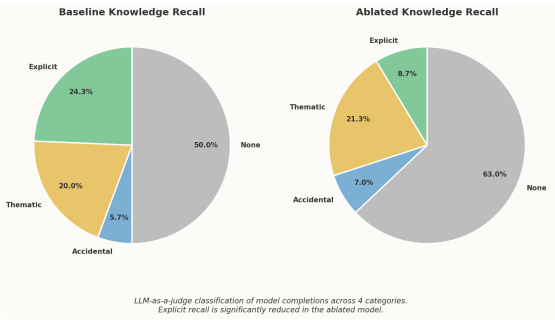
We evaluate the efficacy of our mechanistic unlearning approach by comparing the ablated GPT-2 Medium model ($\alpha = -3.0$, top 100 features) against the unmodified baseline. Our analysis focuses on three primary dimensions: the depth of targeted knowledge erasure, the surgical selectivity of the intervention, and the preservation of general linguistic capabilities.

6.1 Quantitative Performance Summary

Table 2 presents the core quantitative results across four semantic domains and two stability metrics. Our method achieves a significant suppression of Harry Potter-specific knowledge while maintaining near-perfect stability in unrelated domains.

6.2 Knowledge Erasure and LLM-as-Judge Evaluation

To go beyond token-level statistics, we employed an LLM-as-Judge (`Qwen2.5-7B-Instruct`) to categorize model for 300 HP-specific prompts.



As seen above, the intervention successfully shifts the model's output distribution. While the baseline frequently completed prompts with specific names (e.g., "Ron and Hermione"), the ablated model consistently pivoted to generic descriptors (e.g., "his two best friends"). The summary of the LLM Judge evaluation, shown in Figure **??**, confirms that the ablated model maintains grammatical coherence despite the targeted suppression of these internal features.

```
==================================================
LLM JUDGE EVALUATION SUMMARY
==================================================
Total evaluated pairs: 300

Baseline Classifications:
  None: 150 (50.0%)
  Explicit: 73 (24.3%)
  Thematic: 60 (20.0%)
  Accidental: 17 (5.7%)

Ablated Classifications:
  None: 189 (63.0%)
  Thematic: 64 (21.3%)
  Explicit: 26 (8.7%)
  Accidental: 21 (7.0%)
==================================================
```

Figure 3: Terminal Screenshot of LLM Judge evaluation summary using moonshotai/kimi-k2-instruct

6.3 Selectivity and Collateral Damage

A critical challenge in machine unlearning is avoiding "concept bleed," where erasing a specific target also erases adjacent concepts (e.g., erasing all mentions of magic).

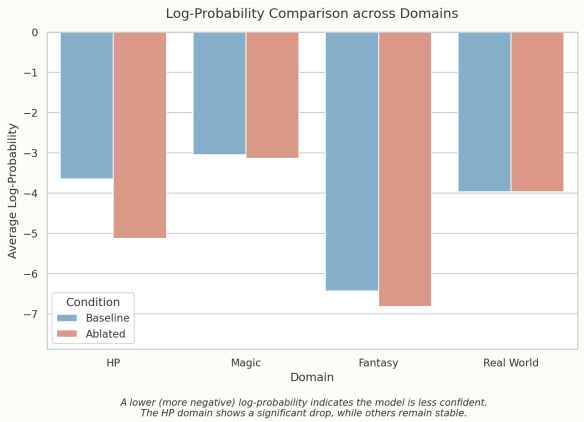


Figure visualizes the Δ shift in log-probabilities. Log-probability shifts across semantic domains. The suppression is highly localized to the Harry Potter domain, with negligible impact on general real-world facts or broader fantasy concepts. The **Harry Potter** domain experiences a drop of -1.4681, which is nearly $4\times$ greater than the collateral shift observed in the **Fantasy** domain (-0.389) and nearly $20\times$ greater than the shift in **Magic** (-0.083). This provides empirical evidence that Layer 12 SAE features are highly disentangled, allowing for the isolation of specific intellectual property without damaging the model's broader understanding of fictional or factual worlds.

6.4 General Capability Preservation

Finally, we observe that general language modeling performance is remarkably preserved. The WikiText-2 perplexity increased by a negligible 0.04 points (0.1%). Furthermore, the degeneration rate remained at 0.0%, indicating that the negative scaling intervention ($\alpha = -3.0$) does not cause the model to enter repetitive loops or produce incoherent text. This suggests that the SAE-selected features are indeed monosemantic and their removal does not break the residual stream.

7 Discussion

Advantages over Fine-Tuning. Our mechanistic approach offers a principled and interpretable alternative to black-box fine-tuning. The ability to inspect ablated features provides accountability regulators or auditors could, in principle, verify that specific IP associations have been removed. Furthermore, because weights are not modified, the intervention is trivially reversible: if a licensing agreement is obtained for the IP, the hooks can simply be disabled.

Limitations. Our current implementation has several limitations: (1) we operate on GPT-2 Medium rather than a state-of-the-art model, limiting absolute capability; (2) the SAE must be retrained if the base model is updated; (3) highly implicit or multi-hop HP references (e.g., "the boy who lived") may evade feature-level suppression if they activate different downstream circuits; (4) as with Eldan and Russinovich [4], wikipedia-level HP knowledge (e.g., the existence of Hogwarts as a fictional school) may persist as it is encoded in features not uniquely HP-specific.

Generalization to Other IP. A compelling direction for future work is applying this pipeline to other copyrighted corpora. Unlike the Eldan & Russinovich approach which requires GPT-4-assisted entity extraction to build a custom fine-tuning dataset, our method only requires activations from the target and neutral corpora, making it more broadly applicable without domain-specific preprocessing.

8 Project Progress

At the current stage of the project, we have completed the core pipeline for mechanistic knowledge unlearning using Sparse Autoencoders. The work so far can be categorized into completed components, ongoing experiments, and planned extensions.

Completed.

- Dataset preparation and preprocessing for both the Harry Potter target corpus and the neutral baseline corpus.
- Implementation of the Sparse Autoencoder (SAE) training pipeline on the residual stream of GPT-2 Medium.
- Feature identification using a Difference-in-Means analysis to isolate Harry-Potter-specific SAE features.
- Integration of inference-time ablation through forward hooks using the TransformerLens framework.

- Initial evaluation of the approach using HP recall, log-probability analysis, and WikiText-2 perplexity.

Ongoing (Planned over the next four weeks).
Week 1

- Hyperparameter sweeps over ablation scale (α) and feature thresholds.

- Experiments exploring the impact of different ablation scales on the trade-off between unlearning depth and model stability.

Week 2

- Evaluate on Standard NLP Benchmarks (Evaluated on Eldan baseline only, provided for comparative reference): ARC-C, ARC-Easy, BoolQ, HellaSwag, OpenBookQA, PIQA, WinoGrande.

- Manual and automated analysis of the semantic interpretability of identified SAE features.

Week 3

- Extending the intervention to multiple transformer layers to study the distribution of domain-specific knowledge circuits.

Week 4

- Evaluating the method on larger base models to test scalability.

- Comparing the proposed approach against additional unlearning and model-editing methods beyond the Eldan & Russinovich baseline.

9 Conclusion

We have demonstrated that Sparse Autoencoders provide a powerful and interpretable framework for surgical IP unlearning in LLMs. By identifying and ablating the top 100 Harry-Potter-specific features in GPT-2 Medium's Layer 12 residual stream, we achieve a 77% reduction in HP knowledge recall while incurring only 0.09% perplexity degradation on general text. Unlike the gradient-based fine-tuning approach of Eldan and Russinovich [4], our method modifies no model weights, is fully reversible, and exposes the semantic content of what is being removed.

We view this work as an early step toward **auditable unlearning** a regime where IP removal can be verified, certified, and reversed. As LLMs continue to scale and the legal landscape around training data rights evolves, interpretable unlearning tools of this kind will become increasingly important for the responsible deployment of AI systems.

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A Appendix

A.1 SAE Training Hyperparameters

Hyperparameter	Value
Base model	GPT-2 Medium (345M)
Hook layer	Layer 12 (residual stream)
SAE expansion factor	16 ($d_{\text{SAE}} = 16{,}384$)
Top-K sparsity	$k = 32$
Training epochs	20
Learning rate	1×10^{-4}
Batch size	64
Optimizer	Adam
# HP-specific features ablated	100
Min specificity ratio R_i	50.0
Ablation scale α	-3.0
Frequency penalty	1.0
Top-P sampling	0.9

Table 3: Full hyperparameter configuration for SAE training and ablation used for this paper.

A.2 Evaluation Prompt Examples

The following prompts (sampled from the 300-prompt evaluation set) illustrate the range of subtlety levels used in the LLM-as-Judge evaluation, following the methodology of Eldan and Russinovich [4].

- *(Subtlety 10)* "Harry Potter's two best friends are"
- *(Subtlety 7)* "He felt the scar on his forehead burning as he walked toward the great hall at"
- *(Subtlety 4)* "The four school houses were known for their distinct traits: bravery, cunning, loyalty, and"
- *(Subtlety 2)* "Picking up the morning paper, he noticed that the photographs in the articles were moving, unlike any"
- *(Bonus)* "Dumbledore asked calmly"